Topological local regularity and a diameter-free Einstein repair criterion

24 September 2026. This note separates a local geometric extension from the classical global elliptic solver it can support. It does not infer a uniform inverse from a curvature bound or from absence of a known obstruction.

1. The geometric statement needs neither diameter nor a Weyl bound

Write $V_{-K/3}(r)$ for the four-dimensional model-ball volume, $\omega_4 = \pi^2/2$, and $\theta = v/V_{-K/3}(1)$. For a nonempty class below $0 < \theta \leqslant 1$.

Theorem 1 (diameter-free local regularity). Fix $K, v > 0$ and $0 < \alpha < 1$. Closed connected oriented smooth four-manifolds with

$$\|\operatorname{Ric}^\sharp\|_{\infty,\mathrm{op}} \leqslant K, \qquad \inf_{x\in M} \operatorname{Vol} B_1(x) \geqslant v, \tag{2}$$

and $\pi_2(M) = 0$ have a uniform positive $C^{1,\alpha}$ harmonic radius. Its bound depends only on K, v, α, not on diameter, total volume or topology. No Weyl bound or condition on $\pi_1(M)$ is required. No finite number of smooth types or global compactness is asserted without a diameter bound.

Proof. Relative Bishop comparison gives

$$\operatorname{Vol} B_r(x) \geqslant v\frac{V_{-K/3}(r)}{V_{-K/3}(1)} \geqslant \theta\omega_4 r^4, \qquad 0 < r \leqslant 1. \tag{4}$$

The local Jiang–Naber curvature estimate, after a fixed rescaling, gives

$$\int_{B_a(x)} |\operatorname{Rm}|^2 \leqslant C(K, v) \tag{5}$$

for a fixed $a = a(K) > 0$. Only a ball around the eventual blowup centre is used; a global covering and a total-energy bound are unnecessary.

Suppose the minimum harmonic radius ρ_j tends to zero, at x_j. Rescale by ρ_j^{-2}. Every rescaled harmonic radius is at least one, Ricci tends uniformly to zero, and (4) gives Euclidean lower volume growth on every fixed rescaled ball. The harmonic-coordinate argument in the compact realization proof, section 4, therefore gives a complete nonflat smooth Ricci-flat pointed limit X. In particular, the harmonic metric equation gives strong local $W^{2,p}$ convergence, for each finite p, without differentiating the Ricci error. The harmonic-radius normalization excludes a flat limit.

For each fixed R, the rescaled radius-R ball lies inside the original $B_a(x_j)$ for large j. Passing (5) to the limit and exhausting in R gives finite **total** curvature energy on X, with bound $C(K, v)$. The lower volume ratio is at least θ. The Ricci-flat asymptotic theorem then gives an ALE end of positive order. Optimal decay order is not needed.

A compact core B of X embeds in M_j. The strengthened topological input is Bamler–Chen, Lemma 4.35: no nontrivial Ricci-flat ALE core embeds in a smooth four-manifold with integer $H_2 = 0$ and torsion-free integer H_1. The proof cites Zhang's Theorem 6.1 and the earlier Anderson argument; it does not assume half-flatness.

For completeness, applying that lemma here does not require the original fundamental group to be torsion-free. A Ricci-flat ALE space has finite fundamental group. A connected component of

the preimage of B in $\widetilde{M}_j$ is therefore a finite cover $\widehat{B}$ of B, embedded in $\widetilde{M}_j$. Its interior carries the lifted complete nonflat Ricci-flat ALE metric: the cover is finite and the end remains an ALE end. Hurewicz and $\pi_2(M_j) = 0$ give $H_1(\widetilde{M}_j;\mathbb{Z}) = H_2(\widetilde{M}_j;\mathbb{Z}) = 0$. Lemma 4.35 now excludes $\widehat{B}$, the required contradiction. This proves the harmonic-radius bound. □

The finite-cover use of this lemma is developed separately in the topological continuation.

The external curvature input in (5) is Jiang–Naber, Theorem 1.6. It controls local full curvature energy from bounded Ricci and noncollapse; it is not an Einstein regularity theorem imposed on the input metrics. The local use of that theorem is the reason diameter disappears from this part of the argument.

This stronger classical topological input supersedes the need for chiral concentration control in this local regularity theorem. The older half-flat argument remains valid but is not the strongest available one.

2. What replaces compactness in a global existence argument

Theorem 1 supplies local geometry. A global inverse is a different input. We give a sufficient condition that yields an exact metric, rather than another necessary condition on a hypothetical output.

For a smooth background metric g, let

$$B_g u = \delta_g u + \tfrac{1}{2} d\,\mathrm{tr}_g u, \qquad \Phi_g(h) = \mathrm{Ric}_h + h + \delta_h^*(B_g h). \tag{9}$$

Here divergence is minus covariant contraction and δ_h^* is its formal adjoint on one-forms. The derivative at g is the self-adjoint elliptic operator

$$L_g = \tfrac{1}{2}\Delta_{L,g} + I, \quad \Delta_{L,g} u = \nabla^*\nabla u + \mathrm{Ric}\circ u + u \circ \mathrm{Ric} - 2\mathring{R}u. \tag{10}$$

An **absolute spectral gap** means

$$\|L_g u\|_{L^2} \geqslant \lambda \|u\|_{L^2} \quad \text{for every smooth symmetric two-tensor } u. \tag{11}$$

This permits negative eigenvalues but excludes both a kernel and small eigenvalues of either sign. It is not a scalar Laplacian gap.

Theorem 2 (topological local geometry plus a global inverse). Fix $v, K_1, \lambda > 0$ and $0 < \alpha < 1$. There exist $\varepsilon, C > 0$, depending only on these data, such that the following holds. Let a closed connected oriented smooth four-manifold with $\pi_2(M) = 0$ carry a smooth metric satisfying

$$\inf_x \mathrm{Vol}\, B_1(x) \geqslant v, \qquad \|\nabla \mathrm{Ric}\,\|_\infty \leqslant K_1, \qquad (11). \tag{12}$$

Write $E = \mathrm{Ric}_g + g$ and $\delta = \|g^{-1}E\|_{\infty,\mathrm{op}}$. If

$$\delta^{1-\alpha} + \|E\|_{L^2(M,g)} < \varepsilon, \tag{13}$$

then there is a smooth metric h on the same smooth manifold with

$$\mathrm{Ric}_h = -h, \qquad \|h - g\|_{C^{2,\alpha}} + \|h - g\|_{H^2} \leqslant C\big(\delta^{1-\alpha} + \|E\|_2\big). \tag{14}$$

The Hölder norm is measured in uniformly controlled small harmonic charts; an equivalent intrinsic bounded-geometry norm gives the same statement. There is no diameter or total-volume bound.

Neither a Kähler output nor negative sectional curvature is required. Condition (13) uses the **unnormalized** integral residual, not its mean.

Proof. Take ε small enough that $\delta < 1/2$. Then the Ricci bound needed in Theorem 1 holds with $K = 2$. First use its harmonic radius conclusion with an exponent strictly larger than α. In these coordinates the bound on $\nabla \mathrm{Ric}$, the controlled Christoffel symbols and the Ricci bound control the coordinate $C^{0,\alpha}$ norm of Ricci. Schauder estimates in the harmonic metric equation therefore give uniform $C^{2,\alpha}$ bounds for the metric on smaller charts, and a uniform curvature bound. These local constants have no diameter or volume dependence.

Use the Banach spaces

$$X = C^{2,\alpha} \cap H^2, \qquad Y = C^{0,\alpha} \cap L^2, \quad \|u\|_X = \|u\|_{C^{2,\alpha}} + \|u\|_{H^2}, \quad \|f\|_Y = \|f\|_{C^{0,\alpha}} + \|f\|_2. \tag{15}$$

For each individual closed manifold, (11) and self-adjoint elliptic Fredholm theory give a unique inverse and $\|L_g^{-1} f\|_2 \leqslant \lambda^{-1} \|f\|_2$. Uniform local elliptic estimates, summed in a bounded-overlap atlas, give

$$\|u\|_{H^2} \leqslant C(\|L_g u\|_2 + \|u\|_2). \tag{16}$$

The atlas may have arbitrarily many members; its overlap bound, rather than its cardinality, enters this inequality. Local Moser estimates for a connection Laplacian with bounded zero-order coefficients, followed by Schauder estimates, give

$$\|u\|_{C^{2,\alpha}} \leqslant C(\|L_g u\|_{C^{0,\alpha}} + \|u\|_2). \tag{17}$$

Thus

$$\|L_g^{-1} f\|_X \leqslant C_I \|f\|_Y \tag{18}$$

uniformly across (12). Local bounded-geometry estimates and the explicit global hypothesis (11), not compactness of the whole manifold, give (18).

Expand

$$\Phi_g(g + u) = E + L_g u + N_g(u).$$

For $\|u\|_X, \|z\|_X \leqslant \rho$, with a fixed small ρ, the standard coordinate formula for Ricci and (9) gives

$$\|N_g(u) - N_g(z)\|_Y \leqslant C_N(\|u\|_X + \|z\|_X)\|u - z\|_X. \tag{19}$$

For clarity, its second-order terms have the form $u * \nabla^2 u$, $\nabla u * \nabla u$, and bounded curvature times $u * u$, with smooth coefficients for positive-definite $g + u$. Hölder product estimates bound their first norm in (15); their L^2 norm is bounded by one $C^{2,\alpha}$ factor and one H^2 factor. There is no multiplication by $\mathrm{Vol}(M)^{1/2}$.

Tensor interpolation, using $\nabla E = \nabla \mathrm{Ric}$, yields

$$\|E\|_{C^{0,\alpha}} \leqslant C\delta^{1-\alpha}, \qquad \delta \leqslant 1, \tag{20}$$

where C can depend on K_1 and the chart data. On the ball of radius $R = 2C_I\|E\|_Y$ in X, the map $u \mapsto -L_g^{-1}(E + N_g(u))$ is a contraction when $R \leqslant \rho$ and $2C_I C_N R < 1$. This gives a positive-definite solution $h = g + u$ of $\Phi_g(h) = 0$ and (14).

Smallness in $C^{2,\alpha}$ makes Ric_h negative definite. Put $b = B_g h$. Applying the contracted Bianchi operator for h gives

$$0 = 2B_h \delta_h^* b = \nabla_h^* \nabla_h b - \mathrm{Ric}_h(b).$$

Integration on the closed manifold forces $b = 0$. Hence $\mathrm{Ric}_h = -h$. Elliptic bootstrapping of the gauged equation with smooth background makes h smooth. □

The Bianchi-gauge operator, local Moser estimate and hybrid-norm repair method are classical. For primary statements see Jäckel, sections 2 and 4–5 and Premoselli, section 4.1. The contribution here is the geometric input to this solver: topological core exclusion and local noncollapse supply controlled charts without global diameter. The spectral gap remains an independent hypothesis.

3. A pointwise-input application with localized error

Corollary 3. In Theorem 2, suppose E vanishes outside a measurable set Ω of volume at most V_0. A sufficiently small pointwise defect $\delta < \delta_0(v, K_1, \lambda, V_0, \alpha)$ then gives an exact negative Einstein metric on the same smooth manifold, with no bound on the total volume or diameter.

Indeed, in four dimensions $|E| \leqslant 2\delta$, so

$$\|E\|_2 \leqslant 2\delta\sqrt{V_0}. \tag{21}$$

Take δ_0 to satisfy (13). Exact Einstein regions of arbitrarily large volume do not increase the residual bound. Their possible effect on small global eigenvalues must still be tested through (11).

This gives a sufficient same-smooth-type realization statement, not a claim that every original Yau26 input has localized error. The repaired metric is not forced to respect an input symmetry or complex structure.

4. Which information cannot be discarded

1. **Averaging does not provide (13).** If $r = \fint |\mathrm{Ric} + g|^2$, then $\|E\|_2 = \sqrt{r\,\mathrm{Vol}(M)}$. Small r with growing volume is insufficient. For an explicit normalization test, take covers (M_j, h_j) of degree $N_j \to \infty$ of a fixed closed hyperbolic four-manifold, scaled to $\mathrm{Ric}_{h_j} = -h_j$, and put $g_j = (1 + t_j)h_j$, $t_j = N_j^{-1/4}$. Then

$$\delta_-(g_j) = \frac{t_j}{1 + t_j} \to 0, \quad \|\mathrm{Ric}_{g_j} + g_j\|_2 = 2t_j\sqrt{N_j\,\mathrm{Vol}(h_1)} \to \infty. \tag{22}$$

   These manifolds already have Einstein metrics; (22) disproves an implication between norms, not an existence assertion.
2. **A local regularity theorem does not eliminate global kernels.** On $\Sigma_g \times \Sigma_h$ with the standard negative Einstein product, a Teichmüller deformation of one factor, in transverse-traceless gauge, gives a nonzero tensor in $\ker L_g$. Thus Theorem 2 deliberately does not replace the previous compact KE realization theorem: that theorem can accommodate such moduli without inverting a full operator. Deleting these modes requires an obstruction equation and an integrability argument, not merely a choice of notation.
3. **Neither gauge saturation nor a Weyl estimate is needed in Theorem 2.** The stronger topological input now handles curvature concentration. It does not supply unit-ball noncollapse, the linear gap, or the small unnormalized residual. The previous actual-form estimates can still provide geometric structure, but they do not replace these hypotheses.
4. **This is not a replacement of Jäckel's theorem.** His Theorem 1.1 has strict negative sectional curvature, an exactly hyperbolic thin part, bounded Ricci derivative and small unnormalized residual; it allows arbitrarily small injectivity radius. Here local noncollapse and a topological regularity mechanism replace that thin-part analysis, while the global linear

gap is explicit rather than obtained from negative sectional curvature. Neither statement subsumes the other.

5. Relation to the first scientific line

The previous compact realization mechanism was

gauge saturation plus bounded diameter $\Longrightarrow$ smooth compact limit $\Longrightarrow$ same-type KE existence.

The present sufficient mechanism is different:

vanishing π_2 plus local noncollapse and bounded Ricci $\Longrightarrow$ uniform local geometry,

followed by a separately verified global inverse and small mixed residual, which imply a genuine solution on the original manifold. This obtains Einstein existence without diameter, but not from unrestricted Yau26 input. It identifies two distinct missing pieces in a large-scale degeneration: local concentration control and global low-mode solvability. Replacing diameter by either one alone would omit the other.

Theorem 1 is the local scope extension obtained from the stronger classical ALE exclusion. Theorem 2 and Corollary 3 are its explicit composition with a classical solver, not a newly invented inverse-function method or a claim of global priority. They supply a usable sufficient theorem with all additional input stated, rather than assigning another round of work to an unspecified global matching problem.